Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 7: Always True, Always False

Tautologies, Contradictions, and Two Rules of Inference

Chartrand & Zhang, Section 1.5

Name: _____ Date: _____

Where we are

Every compound statement you have built so far has been true in some rows and false in others. Today we look at the two extreme cases — always true and always false — and find that the "always true" ones are exactly the rules of reasoning you will use in every proof from Chapter 3 onward. This closes Chapter 1.

By the end of today you should be able to

1. Decide whether a compound statement is a tautology, a contradiction, or neither.
2. State and verify modus ponens and modus tollens.
3. Recognise the two classic fallacies that look like them and are not.

1. Two extremes

Definitions 1.59 and 1.60

A compound statement is a **tautology** if it is true for *all* combinations of truth values of its component statements.
A compound statement is a **contradiction** if it is false for *all* combinations.

The simplest examples use a single component statement.

P	$\sim P$	$P \vee (\sim P)$	$P \wedge (\sim P)$
T			
F			

$P \vee (\sim P)$ is a _____; $P \wedge (\sim P)$ is a _____.

Two facts that follow immediately

(1) S is a tautology if and only if $\sim S$ is a _____.
(2) If $R \equiv S$, then $R \Leftrightarrow S$ is a _____. Every equivalence you proved in Weeks 1 and 2 was secretly a tautology.

With an open sentence. Let $P(n)$: n is even, over the integers.

- $P(n) \vee (\sim P(n))$: _____ — true for every integer.
- $P(n) \wedge (\sim P(n))$: _____ — false for every integer.

2. Classify these

P	Q	$P \Rightarrow Q$	$\sim Q$	$(\sim Q) \vee (P \Rightarrow Q)$	$(P \wedge Q) \wedge (Q \Rightarrow (\sim P))$
T	T				
T	F				
F	T				
F	F				

Column five is a _____. Column six is a _____.

3. Modus ponens — the rule you already use

Modus ponens

$$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$$

If P is true, and P implies Q, then Q is true.

P	Q	$P \Rightarrow Q$	$P \wedge (P \Rightarrow Q)$	$(P \wedge (P \Rightarrow Q)) \Rightarrow Q$
T	T			
T	F			
F	T			
F	F			

True in every row, so it is a _____.

4. Modus tollens — the contrapositive at work

Modus tollens

$$((P \Rightarrow Q) \wedge (\sim Q)) \Rightarrow (\sim P)$$

If P implies Q, and Q is false, then P is false.

This is also a tautology — verify it on your own paper.

You have already used both

Quiz 2 asked: the requirement "if a file is encrypted, then it is backed up" holds, and a file is *not* backed up. Concluding the file is not encrypted is exactly _____.
These two tautologies are the engine of every proof in Chapter 3. Modus ponens drives direct proof; modus tollens drives proof by contrapositive.

5. Two impostors

Change one symbol in each rule and it stops being a tautology.

P	Q	$(Q \wedge (P \Rightarrow Q)) \Rightarrow P$	$((P \Rightarrow Q) \wedge (\sim P)) \Rightarrow (\sim Q)$
T	T		
T	F		
F	T		
F	F		

Neither is a tautology. Both fail in the same row: P _____, Q _____.

These have names, and you have met them before

The first is **affirming the consequent**: knowing Q and $P \Rightarrow Q$, concluding P. That is assuming the *converse*.
The second is **denying the antecedent**: knowing $\sim P$ and $P \Rightarrow Q$, concluding $\sim Q$. That is assuming the *inverse*.
Week 2 showed converse and inverse are equivalent to each other and *not* to the original. That is precisely why both fallacies fail, and why they fail in the same row.

Which rule, if any? Mark each argument **valid** or **fallacy**.

Argument	Verdict
If the tests fail, the build stops. The tests failed. So the build stopped.	
If the tests fail, the build stops. The build stopped. So the tests failed.	
If the tests fail, the build stops. The build did not stop. So the tests did not fail.	

Exit ticket

1. Is $(P \Rightarrow Q) \vee (Q \Rightarrow P)$ a tautology, a contradiction, or neither? One line of justification.

2. Name the rule: "If it rained, the ground is wet. The ground is not wet. So it did not rain."

Before Wednesday

Read *Mathematics for Computer Science* (Lehman, Leighton, and Meyer), Section 3.4.1, "Propositions in Normal Form," pp. 50–52. The link is on Canvas.

Note: Chartrand does not cover normal forms, so Wednesday and Friday use the free supplement instead. Everything else this term stays in Chartrand.

Self-check (not collected). Section 1.5, Exercises 1, 3, 5, 7, and 9 — all odd-numbered, so answers are in the back. Exercise 5 is modus tollens.

Coming up

Homework 3 is due Friday, September 10, 11:59 p.m.
Wednesday and Friday cover conjunctive and disjunctive normal form — the last topics in our logic unit.

Questions I still have